# CCCA Cell-State Analysis — Filbin et al. 2018

**Pediatric H3-K27M Mutant Glioma · Brain**

**Dataset.** H3-K27M diffuse midline glioma · SmartSeq2 · 2,118 malignant cells, 5 samples (log1p TPM).

## GO / NO-GO GATES

**Gate 1 · Sample mixing + continuity** — **GO**

– Harmony θ-sweep: sample iLISI 1.18→3.56 of 5, plateaus by θ=5–10, robust across n_pcs 20/30/40.
– One connected component (0 detached islands); elongated single cloud (aspect 1.83). Mixed + continuous.

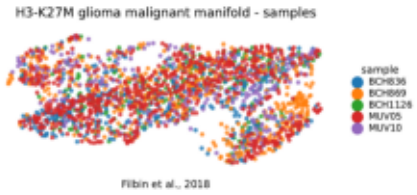
**Gate 2 · Embryonic module geography** — **GO**

– Gast/pSom co-vary (r=+0.56) over a shared body; both oppose Crest (r=−0.87, −0.57).
– Crest is an inverse terminal gradient (crest-high decile at pseudotime 0.99). Non-branching pattern present.

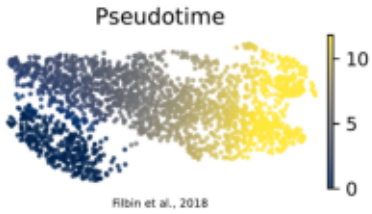
**Gate 3 · Gast↔pSom cross-repression** — **NA**

– No competing arms: Gast and pSom co-vary rather than diverge — no bifurcation hub to test.
– Branching-only correlation/modules steps not produced (per §6).

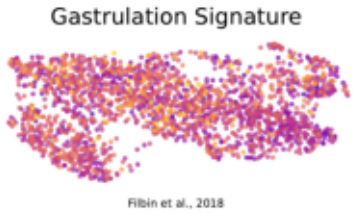
## STEP FIGURES



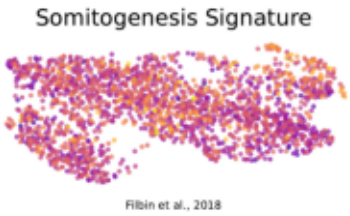

01 · Integration (samples)



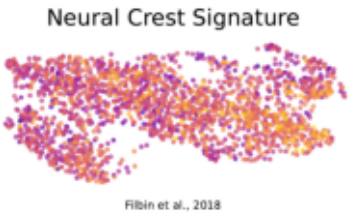

02 · Trajectory (pseudotime)



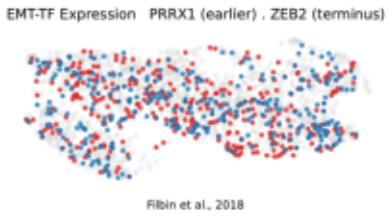

03 · Gastrulation




03 · Somitogenesis




03 · Neural Crest




04 · EMT-TFs

## KEY PARAMETER CHOICES

| | |
|---|---|
| **Min-cell threshold:** | ≥184 (geometric mean across largest count gap 241→141); kept 5/6 samples → 2,118 cells. |
| **Integration / embedding:** | Harmony θ=10 on sample; n_pcs=20; neighbors k=30; UMAP spread=1.3; no sc.pp.scale. |
| **Geometry:** | Linear — PPT tree fragments to 36–54 forks (a bush); ElPiGraph curve is 2-tip/0-fork, fold-free (2.49). |
| **Root:** | Curve tip highest in Gast+pSom, lowest in Crest (origin); continuity DC1 ρ=+0.93 (independent). |
| **EMT-TFs shown:** | PRRX1 (earlier) · ZEB2 (terminus) — detection-gated (frac>0.03), 10-TF mesenchymal panel. |

**Verdict.** Pattern reproduces (non-branching): one continuous gastrulation/somitogenesis → neural-crest axis (G1 GO · G2 GO · G3 NA); no Gast/pSom arm split. Matches the Brain-category norm.